

Task 2: React Frontend for Task Manager Application

Objective

Develop a React-based frontend for the Task Manager Application that allows users to interact with the backend API through a user-friendly interface.

Technologies Used

- React.js
- JavaScript
- HTML5
- CSS3
- Axios
- Node.js

Project Overview

The frontend application provides an interface for managing tasks. Users can:

- View all tasks
- Add new tasks
- Update task details
- Delete tasks
- Communicate with the backend API

The application was built as a Single Page Application (SPA) using React.

Features Implemented

1. Display Tasks

- Fetches task data from the backend API.
- Displays tasks in a structured format.

2. Add New Task

- Form provided to enter task details.
- Sends POST request to backend API.

3. Update Task

- Allows modification of existing task information.
- Sends PUT request to backend API.

4. Delete Task

- Removes selected task.
- Sends DELETE request to backend API.

5. API Integration

- Axios used for communication with backend services.
- Handles API responses and updates UI dynamically.

Folder Structure

task-manager-frontend/

├── public/

├── src/

| ├── App.js

| ├── index.js

| ├── components/

| └── styles/

├── package.json

└── README.md

Implementation Steps

Step 1: Create React Application

`npx create-react-app task-manager-frontend`

Step 2: Install Axios

npm install axios

Step 3: Build Components

- Task List Component
- Add Task Form
- Edit Task Form

Step 4: Connect with Backend

Configured API endpoints using Axios for CRUD operations.

Step 5: Run Application

npm start

Frontend runs on:

<http://localhost:3000>

Testing

The following functionalities were tested successfully:

- Create Task
- Read Tasks
- Update Task
- Delete Task
- API Communication

All operations worked correctly and reflected changes in the user interface.

Outcome

Successfully developed a React frontend application for the Task Manager system. The frontend communicates with the backend API and supports complete CRUD operations through an intuitive user interface.

Github

<https://github.com/afrosefathima455-hub/task-manager-api.git>